

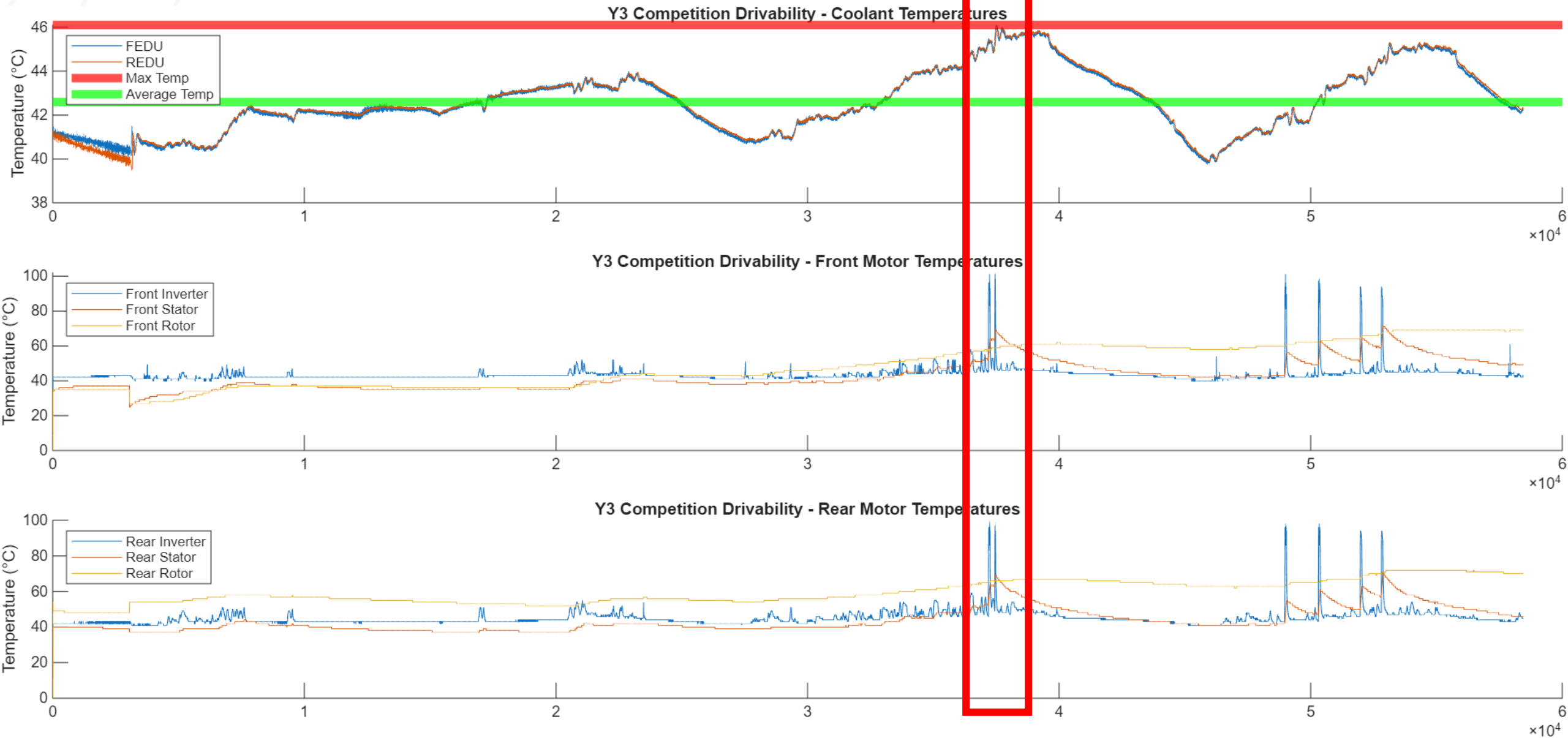
Heat Exchanger Selection & Mounting

Hand Calculations & FEA

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- Costs/Advantages
- Coolant Temp
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Problem - Overheating

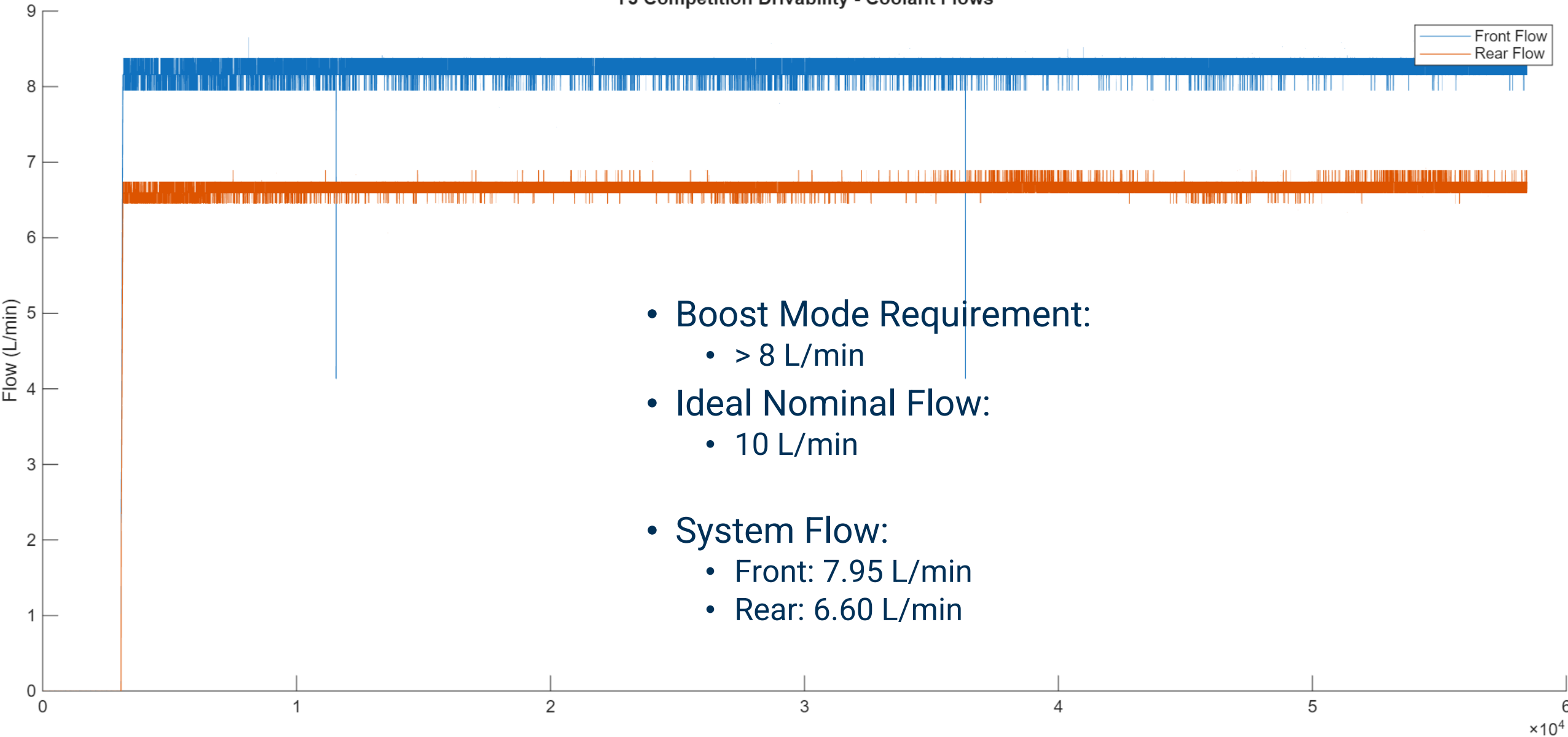


Problem - Overheating

- This issue is manageable in software (we can add a hysteresis so the torque doesn't oscillate so much)
- The actual temperature limit is different than what Magna originally indicated
 - Was 105°C -> Now 130°C
- Increasing the flow should completely remove this issue

Problem – Boost Mode Inoperable

Y3 Competition Drivability - Coolant Flows



- Boost Mode Requirement:
 - > 8 L/min
- Ideal Nominal Flow:
 - 10 L/min
- System Flow:
 - Front: 7.95 L/min
 - Rear: 6.60 L/min

Costs/Advantages

Current System

- Cost: \$0
- Points: -45/1000 (for 0-60 and 50-70 acceleration)
- Time: 0
- Risk: No boost mode or maximum performance

Modified System

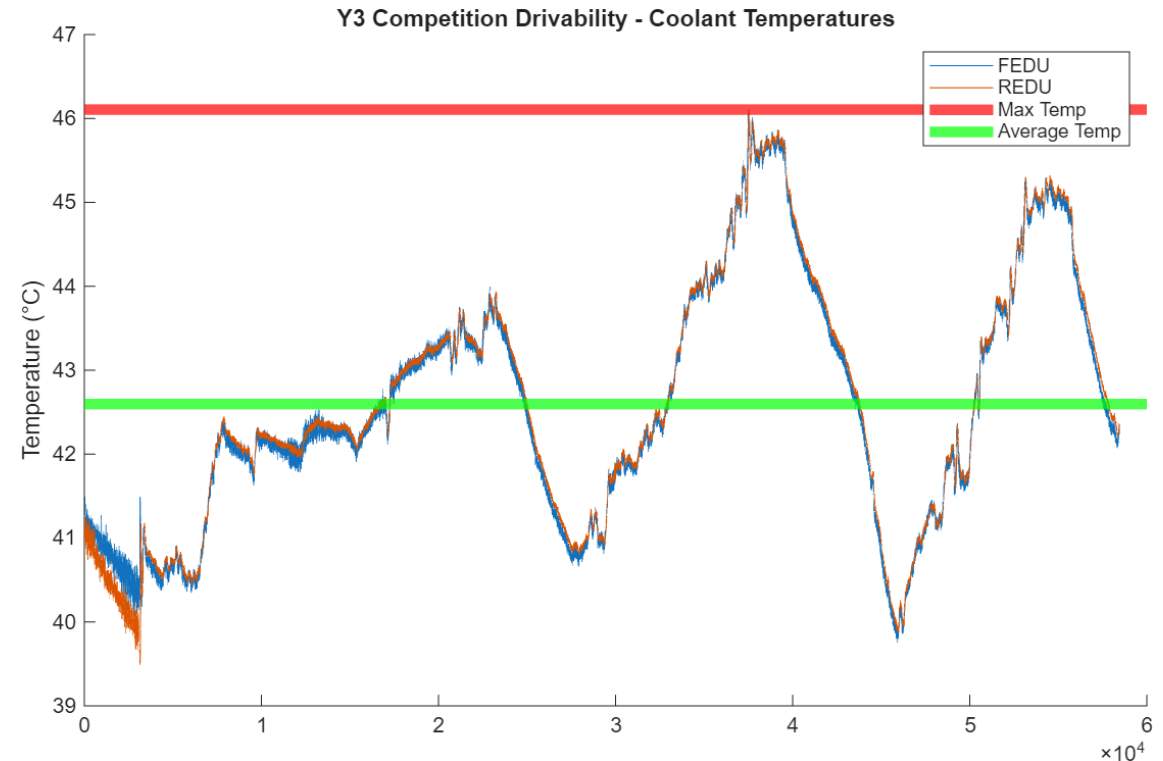
- Cost: \$1k-3k
- Points: +45/1000 (for 0-60 and 50-70 acceleration)
- Time:
 - 60+ hrs design
 - 30+ hrs manufacturing
- Risk:
 - Vehicle not ready for CARB
 - Motors overheat **more**

Coolant Temp

- Max Coolant Temp = 46.11 °C
- Nominal = 42.6 °C
 - Note: Max coolant temp during CARB PCM energy consumption event = 36.5 °C

This data proves the radiator is sufficient for thermal rejection of system

- Max Allowable Coolant Temp = 57 °C
 - Magna EDU requirement



Flowrate

- Current:
 - Front: 7.95 L/min
 - Rear: 6.60 L/min
- Desired:
 - Front: 10 L/min
 - Rear: 10 L/min
 - *Nominal (max)*
 - 20 (25) L/min
 - 5.28 (6.60) GPM

Heat Rejection

- Maximum
 - 91,746 BTU/Hr
 - 26.888 kW
- Nominal
 - 25,796 BTU/Hr
 - 7.56 kW

Heat Rejection - Power

- Power = Work / Time

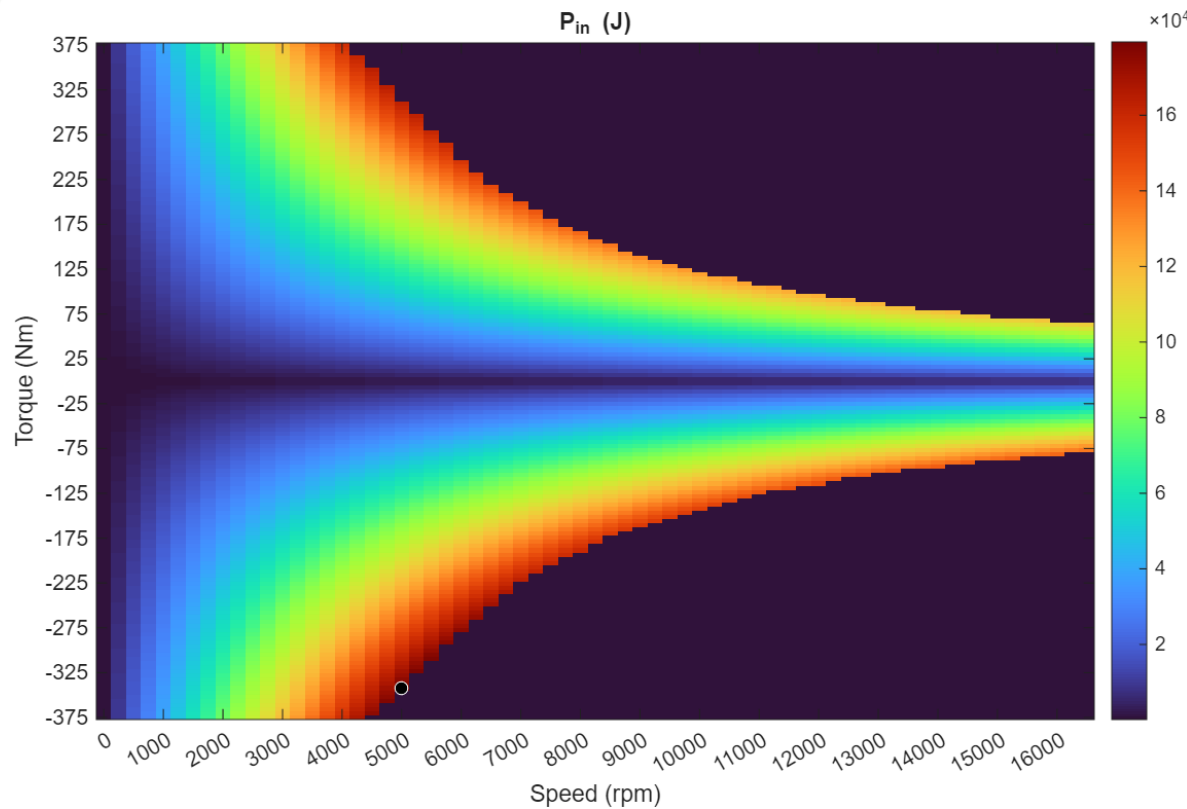
- $P = \frac{W[J]}{t[s]} = \frac{F[N]r[m]}{t[s]} = \tau[Nm] * \omega \left[\frac{rad}{s} \right]$

- $P = \tau[Nm] * rpm \left[\frac{rev}{min} \right] * \frac{2\pi}{60}$

- $J = \frac{kgm^2}{s^2} = Nm = \frac{kgm}{s^2} * m$

- $N = \frac{kgm}{s^2}$

- $\omega \left[\frac{rad}{s} \right] = rpm \left[\frac{rev}{min} \right] * \frac{1min}{60s} * \frac{2\pi rad}{1rev}$



Heat Rejection - Efficiency

- $\eta = \frac{P_{out}}{P_{in}} * 100\%$
- $P_{out} = \frac{\eta}{100} P_{in}$
- $P_{in} = P_{out} + P_{loss}$
- $P_{loss} = P_{in} - P_{out}$

Confidential Efficiency Map for the Electric Drive Units

Heat Rejection - Power Loss

- $P_{loss} = P_{in} - P_{out} = P_{in} - \frac{\eta}{100} P_{in} = P_{in} \left(1 - \frac{\eta}{100}\right) = \tau * rpm * \frac{2\pi}{60} \left(1 - \frac{\eta}{100}\right)$

- Maximum: 13.444 kW

- Average: 3.780 kW



- Two Motors:

- Maximum: 26.888 kW

- Average: 7.56 kW

Confidential Power Loss Map for the Electric Drive Units

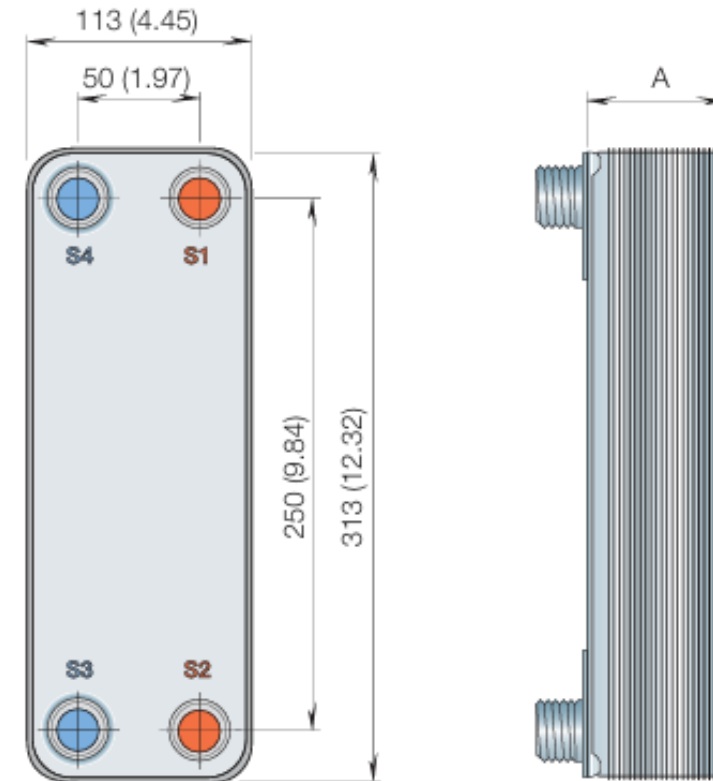
Heat Exchanger

Selected

- Max Pressure: 3100 kPa
- Max Temperature: 204 °C
- Weight: 6.97 lb
- Cost: \$ 564.51
- Total Area: 0.636 m²
- Pressure Drop: 7 kPa
- Max Flow: 62 GPM
- [CB30-18H - Alfa Laval CB30-18H - 450,000 BTU/Hr \(Domestic Water\) CB30 Braze Plate Heat Exchanger \(18 Plates\)](#)

For more information, see the following slides in this section which cover alternatives and more of the reasoning.

Measurements in mm (inches)



Heat Exchanger

Wessels

- Water - Water
- 125,000 > 91,746 BTU/Hr
- 5.2 ~ 5.28 GPM
- Weight: 5 lb
- Cost: ?

[How To: Size Braze Plate Heat Exchangers | Wessels Company](#)

DOMESTIC HEATING w 180°F – DOUBLE WALL					
DUTY BTU/HR	MODEL	BOILER		DOMESTIC	
		180°F to 130°F		50°F to 140°F	
		GPM	ΔPSI	GPM	ΔPSI
25,000	WB15D-10	1.0	0.3	0.6	0.1
50,000	WB15D-10	2.1	1.0	1.1	0.2
75,000	WB15D-10	3.1	2.1	1.7	0.5
100,000	WB15D-10	4.1	3.6	2.2	0.8
125,000	WB15D-10	5.2	5.5	2.8	1.2
150,000	WB15D-10	6.2	7.7	3.3	1.7
175,000	WB15D-20	7.2	2.2	3.9	0.6
200,000	WB15D-20	8.3	2.9	4.4	0.8
225,000	WB15D-20	9.3	3.6	5.0	1.0
250,000	WB15D-20	10.3	4.5	5.6	1.3
275,000	WB15D-20	11.4	5.4	6.1	1.5
300,000	WB15D-20	12.4	6.3	6.7	1.8
350,000	WB15D-20	14.4	8.5	7.8	2.4
400,000	WB15D-40	16.5	5.0	8.9	1.5
450,000	WB15D-40	18.6	5.2	10.0	1.6
500,000	WB15D-40	20.6	5.7	11.1	2.0
600,000	WB15D-40	24.8	6.6	13.3	1.7
700,000	WB15D-40	28.9	7.9	15.6	2.5
800,000	WB15D-50	33.0	7.5	17.8	2.4
900,000	WB15D-60	37.1	7.5	20.0	2.1
1,000,000	WB15D-70	41.3	7.1	22.2	2.1
1,100,000	WB15D-80	45.4	7.2	24.4	2.2
1,200,000	WB15D-90	49.5	7.5	26.7	2.3
1,300,000	WB15D-100	52.0	7.3	30.0	2.3

Heat Exchanger

Wessels

- Water - Water
- 100,000 > 91,746 BTU/Hr
- 5.0 ~ 5.28 GPM
- Weight: 5 lb
- Cost: ?

[How To: Size Braze Plate Heat Exchangers | Wessels Company](#)

DOMESTIC HEATING w 140°F – DOUBLE WALL					
DUTY BTU/HR	MODEL	BOILER		DOMESTIC	
		140°F to 100°F		50°F to 120°F	
		GPM	ΔPSI	GPM	ΔPSI
25,000	WB15D-10	1.3	0.1	0.7	0.1
50,000	WB15D-10	2.5	1.5	1.4	0.3
75,000	WB15D-20	3.8	0.7	2.1	0.2
100,000	WB15D-20	5.0	1.2	2.9	0.4
125,000	WB15D-20	6.3	1.9	3.6	0.5
150,000	WB15D-20	7.5	2.6	4.3	0.8
175,000	WB15D-40	8.8	1.3	5.0	0.4
200,000	WB15D-40	10.0	2.8	5.7	0.6
225,000	WB15D-40	11.3	2.3	6.4	0.7
250,000	WB15D-40	12.5	2.9	7.1	0.9
275,000	WB15D-40	13.8	2.3	7.9	0.7
300,000	WB15D-40	15.0	2.7	8.6	0.9
350,000	WB15D-40	17.5	3.7	10.0	1.2
400,000	WB15D-40	20.0	9.0	11.4	2.3
450,000	WB15D-50	22.5	3.9	12.9	1.3
500,000	WB15D-60	25.0	3.5	14.3	1.2
600,000	WB15D-70	30.0	4.2	17.1	1.4
700,000	WB15D-80	35.0	4.6	20.0	1.5
800,000	WB15D-90	40.0	5.2	22.9	1.7
900,000	WB15D-100	45.0	6.0	25.8	2.0
1,000,000	WB15D-120	50.0	7.2	30.0	2.4

Heat Exchanger

Alfa Laval

[New selection*](#)

We now know the stock side should have a flow of 11.3L/min and to assume a nominal input temperature on the cold side of 20°C

Current folder*

Default Folder

Selection name*

New selection

Update

Add

1 Sizing

2 Results

SI/Second

Calculation input

Capacity (kW)

26.89

Flow type (liquid)

☐ Mass

☒ Volume

Max Heat Transfer

Max Flow Rate

Min Outlet Temperature

Hot side

Inlet temperature

76.64 °C

Fluid

Water

Flow rate

0.3333 l/s

Max pressure drop

70.0 kPa

Outlet temperature

57.00 °C

Cold side

Outlet temperature

65.66 °C

Fluid

Water

Flow rate

0.33 l/s

Max pressure drop

70.0 kPa

Inlet temperature

46.11 °C

Max Flow Rate

Max Inlet Temperature

☒ Selection ☐ Rating

Region

North America

Product model

Product input

Calculate result list

Heat Exchanger

Danfoss

[Danfoss HEXSelector | Danfoss](#)

HEX Type

BHEX HVAC / HVAC 1.2

Basic

Duty Type

Liquid

Calculation Mode

Design

Calculation Options

Heat Exchanger Configuration

Heat Exchanger

Auto

Filter

Channel Type

Auto

Duty

Side 1

Side 2

Load

kW

26.89

Minimum Margin

%

0

Fluid

Ethylene glycc

Ethylene glyco

Temp Limits:

-35.5 / 197.3°C

-35.5 / 197.3°C

Concentration

%

50.0050.00

Inlet Temperature

°C

79.0646.11

Outlet Temperature

°C

57.0068.30

NTU1/NTU2/LMTD(K): 2.04 / 2.05 /

Flow Rate

20.00

L/mi

L/mi

20.00

Max Pressure Drop

kPa

200.00200.00

Real OutCalculate

Calculation Default

Make Default

Border Style

Fixed3D

Heat Exchanger	Units In	Margin	Number of Plates	Grouping	Passes	Total Area (m²)
XB05H-1-20	1	6.5	20	1*9H/1*10H	1	0.38
XB06H+-1-16	1	2.5	16	1*7H+/1*8H+	1	0.38
XB05M-1-30	1	3.1	30	1*14M/1*15M	1	0.59
XB12H-1-20	1	0.6	20	1*9H/1*10H	1	0.50
XB37H-1-10	1	29.7	10	1*4H/1*5H	1	0.45
XB06H-1-36	1	7.5	36	1*17H/1*18H	1	0.92
XB12M-1-30	1	1.1	30	1*14M/1*15M	1	0.78
XB37M-1-16	1	25.0	16	1*7M/1*8M	1	0.78
XB37L-1-16	1	6.5	16	1*7L/1*8L	1	0.78
SL70TL-1-10	1	23.1	10	1*4TL/1*5TL	1	0.58
XB05X-1-60	1	5.0	60	1*30X/1*29X	1	1.22
XB06L-1-50	1	6.6	50	1*24L/1*25L	1	1.30
XB12L-1-50	1	2.9	50	1*24L/1*25L	1	1.34
XB12M-2-20/20	1	79.6	40	(1*9M+1*10M)/(1*10M+1*10M)	2	1.06
XB12H-2-20/20	1	108.6	40	(1*9H+1*10H)/(1*10H+1*10H)	2	1.06

Heat Exchanger

XB05H-1-20

Danfoss Code

079G1442

Water quality

Show 1

Duty & Media

Dimensional Data

Accessories

Punch List

Messages

Calculated Parameters

Unit

Side 1

Flow Type

CounterCurrent

Heat Load

kW

26.89

Inlet Temperature

°C

79.1

Outlet Temperature

°C

57.0

Mass Flow Rate

kg/s

0.35

Volumetric Flow Rate

L/min

20.00

Total Pressure Drop

kPa

117.40

Pressure Drop in Port

kPa

0.81

Effective Area

m²

0.38

Surface Margin

%

6.5

Heat Exchanger Comparison

[Heat Exchangers.xlsx](#)

		Max Heat Exchanged and Flow and Pressure Drop?	Heat Exchanged (BTU/Hr)	Heat Exchanged (kW)	Flow1 (GPM)	Flow2 (GPM)	Pressure Drop 1 (PSI)	Pressure Drop 1 (kPa)	Pressure Drop 2 (PSI)	Pressure Drop 2 (kPa)	A	B	C	D	E	F	Volume (Gals/Hr)	Area Per Channel (ft²)	Total Area (ft²)	Plates	Cost (\$)	Weight (lb)	Coverage Factor (%)	High	Notes	
Brand	Number																									
Bell & Gossett	BP400-10 (3/4" MPT)	Yes	99,900.00	29.28	4.10	5.40	4.10	28.26	5.00	34.46	9.84	12.20	1.97	4.40	0.95	0.09	0.01	0.13	1.26	10.00	386.19	3.00	0.09		Good option. The low area is likely made up in the high pressure drop	
Bell & Gossett	BP415-60 (1" MPT)	Yes	97,900.00	28.79	3.41	4.34	0.32	2.21	0.21	1.43	16.33	20.76	1.97	4.40	0.95	0.09	0.07	33.98	80.00	9,337.93	74.89	0.04			Bad. Too big because focus was on small pressure drop	
Wessels	WB15D-10	Yes	125,000.00	36.63	5.20	2.80	5.50	37.90	1.20	8.27	9.84	12.05	2.76	4.88	1.39			0.34	3.40	10.00			8.00	0.36		Can't find where to buy
Wessels	WB15D-20	Yes	100,000.00	29.31	5.00	2.90	1.20	8.27	0.40	2.76	9.84	12.05	2.76	4.88	2.29			0.34	6.80	20.00			11.00	0.09		Can't find where to buy
Alfa Laval	CB110AQ-16L	No	91,745.66	26.89	5.28	5.28	0.07	0.46	0.09	0.60	36.02	24.25	3.62	7.52	1.89	2.20		1.27	20.26	16.00	1,714.01	27.47	2.20	CB110-16L	Too expensive	
Alfa Laval	CB60AQ-20H	No	91,745.66	26.89	5.28	5.28	1.22	8.42	1.51	10.41	18.35	20.75	0.20	4.45	0.94	2.22		0.64	12.82	20.00	802.61	12.76	105.50	CB60-20H	Coverage factor is big (it is too overkill) and price is larger	
Alfa Laval	CB30AQ-18H	No	91,745.66	26.89	5.28	5.28	0.80	5.49	1.00	6.92	9.84	12.32	1.97	4.45	0.94	2.17		0.38	6.85	18.00	564.51	7.74	4.60	CB30-18H	Good option	
Alfa Laval	AlfaNova 27-20H	No	91,745.66	26.89	5.28	5.28	1.86	12.80	2.29	15.80	18.35	20.71	1.97	4.37	0.94	2.40		0.63	12.57	20.00	2,521.99	14.26			Too expensive	
Alfa Laval	AlfaNova 52-20H	No	91,745.66	26.89	5.28	5.28	1.86	12.80	2.29	15.80	18.35	20.71	1.97	4.37	0.94	2.40		0.63	12.57	20.00	2,521.99	14.26			Alfa Laval Too expensive	
Alfa Laval	AlfaNova 76-20H	No	91,745.66	26.89	5.28	5.28	0.26	1.79	0.32	2.22	20.43	24.33	3.62	7.52	1.89	2.68		1.27	25.41	20.00	4,629.99	36.16			Alfa Laval Too expensive	
Danfoss	XB05H-1-20	No	91745.66429	26.888	5.28	5.28	17.927491	127.4	17.927491	117.4	10.343	12.203	1.834	2.992	0.707	1.010	0.00423	4.88	20.00			3.22	0.30		Pressure drop too much	
Danfoss	XB12H-1-20	No	91745.66429	26.888	5.28	5.28	2.8441901	19.61	2.3452603	16.17	9.2126	11.378	2.48	0.709	0.787	1.339		0.06	5.38	20.00	423.89	6.90	0.60	004H7688	Good option	
Danfoss	XB37H-1-10	No	91745.66429	26.888	5.28	5.28	11.339051	78.18	7.8204349	53.92	18.858	20.669	2.835	4.685	0.787	0.965		0.67	4.84	10.00	489.25	9.26	29.70	004H7300	Pressure drop too much	

Some HXs stated their max operating values and other stated what was imported into a calculator

Total area x pressure should be proportional to the heat transfer.

- 1) Pressure drop: 30 kPa
- 2) Pressure drop: 7 kPa
- 3) Pressure drop: 20 kPa

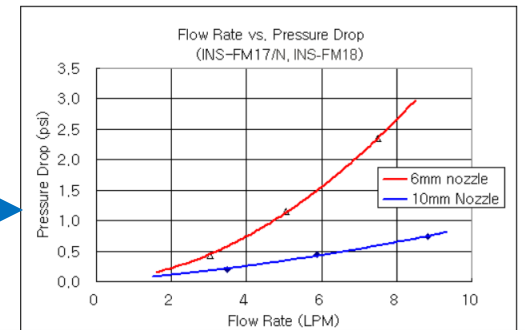
** The heat transfer of 26.888 kW is considerable conservative on its own.

As turbulence increases, so does the pressure drop and heat transfer. I expect this means that as pressure drop increases, so does the heat transfer.

System Architecture: Current

Assumptions:

- 0% LTR
- Temp
 - 60 °C (GM Parts)
 - 20 °C (FDU + RDU)this gives us a conservative estimate; as temperature increases pressure decreases
- Flow (Ideal)
 - 20 L/min (system)
 - 10 L/min (FDU)
 - 10 L/min (RDU)



Confidential Thermal Architecture and Pressure Drops

* The raised IPE should not add significant resistance to the system. Whatever pressure drop occurs when the coolant travels up will be added back once the coolant moves back down.

Source of Pressure Drops: [Magna Pressure Drop PowerPoint](#) – Linearly interpolated from 9-12 L/min (I don't think I can share a copy because it is confidential? Need access to GT SharePoint)

System Architecture: Pressure Requirements

The HX effectively replaces the stock drive units (which are in parallel) and OBCM2. See the following slide for more detail.

- Stock:

- Drive Units
- OBCM2: ~
- Total: ~44.8453 kPa

Confidential
Pressure
Drops

This means if the added components have a pressure drop of less than 44.8 kPa, the flow should be 20 L/min.

This is not the case for the current system, which appears to have a pressure drop of about 31.4 kPa. Either the pressure needs to be under 31.4 kPa, or the current system has a larger pressure drop than 31.4 kPa due to additional bends.

Each 90 degree connector can add a pressure drop of about 0.24 kPa (see [here](#) for more detail). It is conceivable there are 10 bends which would contribute a total of 2.4 kPa of pressure drop.

System Information: MY24 AWD LYRIQ

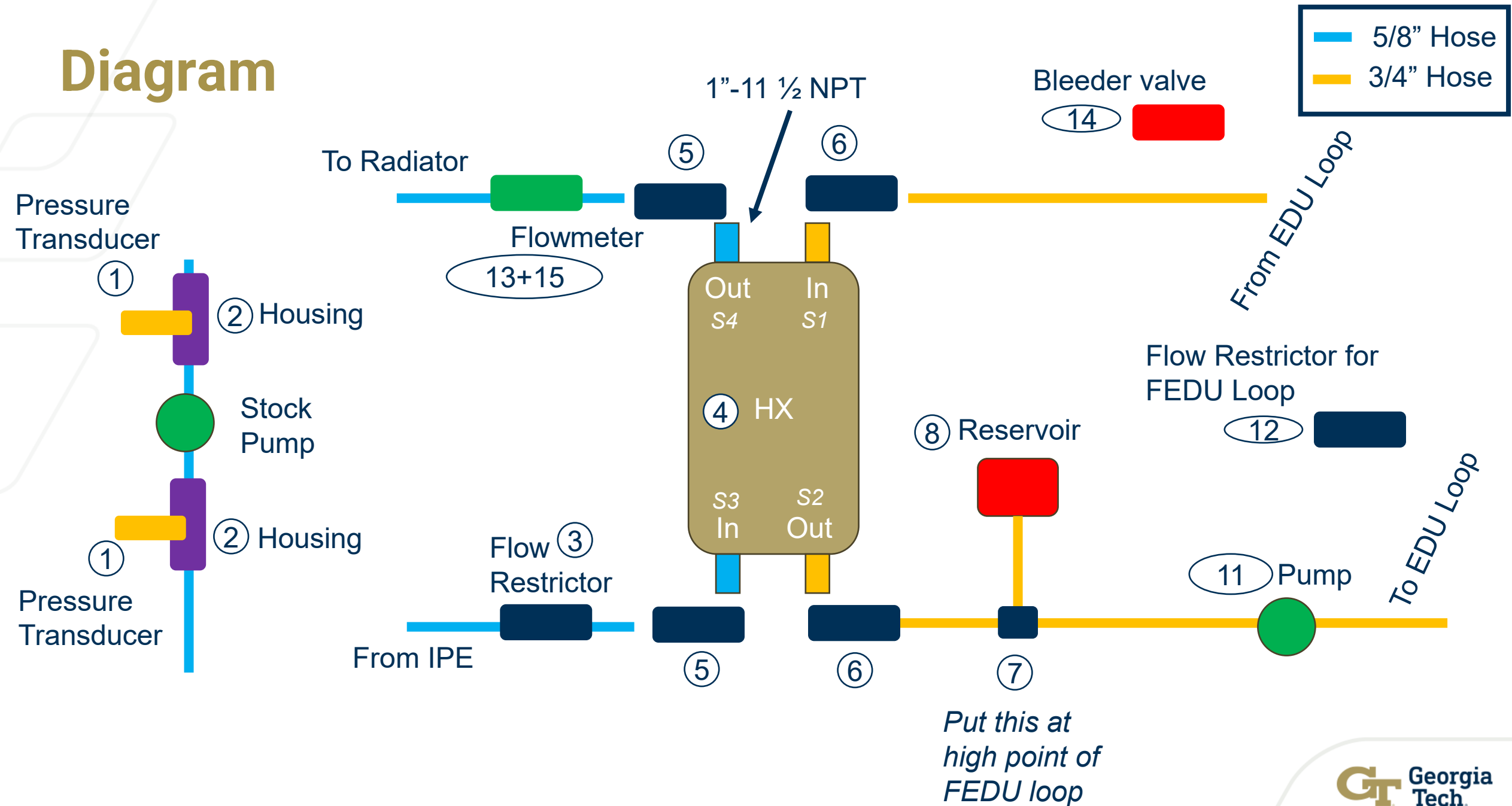
Assumptions:

- 0% LTR
- 60 °C
- Flow
 - 20.4 L/min (system)
 - 10.1 L/min (FDU)
 - 10.3 L/min (RDU)

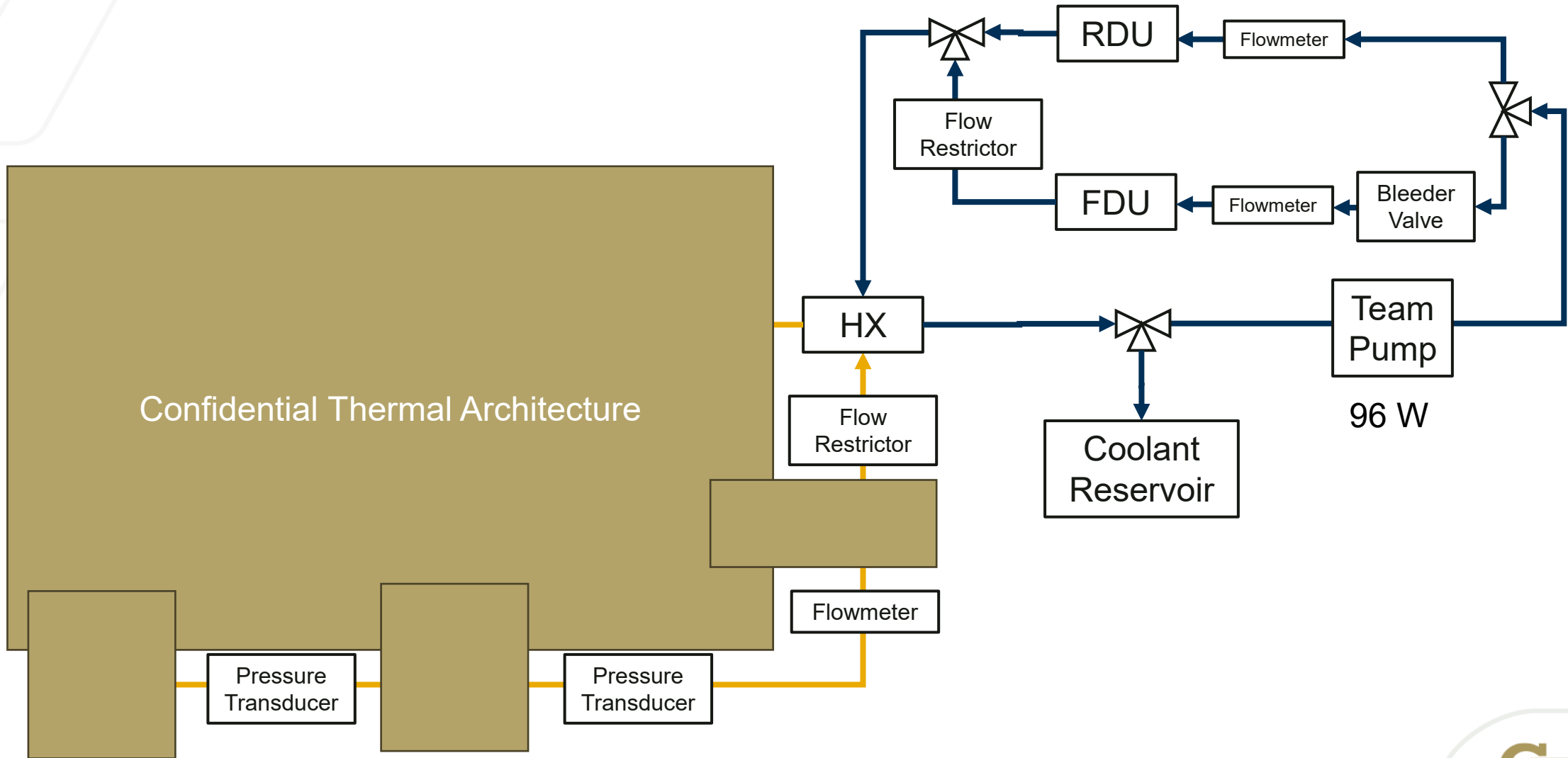
Confidential Thermal Architecture and Pressure Drops

Source of Pressure Drops: GM Garbage Collector V3 <https://anl.box.com/s/b6m2rktyqaqkx6ymi9irjzm4hzs1ls4f>
(Need EcoCAR BOX Access)

Diagram



System Architecture: Proposed



BOM

https://gtvault.sharepoint.com/:x/s/ECE/EC-oCAR/IQDrzrVUKhcvTZdulMjvDLMzAQ_aPXZBBRcs_65Yqt2Rs8o?e=LRCEgL

Number in Diagram	Quantity	Part #	Description	Price Per Unit	Total Cost	Link	Notes
4	1	CB30-18H	Alfa Laval Brazed Plate Heat Exchanger	\$564.51	\$564.51	https://www.supplyhouse.com/Alfa-Laval-CB30-18H-450000-BTU-Hr-Domestic-Water-CB30-Brazed-Plate-Heat-Exchanger	
2	2		Glow Shift: 16mm (5/8") Water Sender Hose	\$22.99	\$45.98	https://www.glowshiftdirect.com/16mm-5-8-water-sender-hose-attachment/	
1	2		DATAQ 0-60 PSI high accuracy pressure transducer	\$68.00	\$136.00	https://www.dataq.com/products/accessories/pressure-sensor/2000361-hs-60.html	
3	1	84706	5/8" Manual Adjustment Heater Valve	\$11.99	\$11.99	https://www.summitracing.com/parts/fss-84706	
6	1	(Pack of 2)	3/4" Barb to 1" NPT Female Fitting (Brass)	\$15.49	\$15.49	Kozelo Pack of 2 Brass Hose Barb Fittings Adapter	
5	2		5/8" Barb to 1" NPT Female Fitting (Stainless Steel)	\$9.99	\$19.98	HARFINGTON 1pcs 5/8" to 1" NPT Female Sta	Depends on waiver; wait to order
7	1		3/4" x 3/4" x 5/16" Tee Hose Adapter	\$13.33	\$13.33	Amazon.com: beler 3/4" 5/16" 3/4" Tee Heater Hose Coolant Reducer Fitting Water Connector : Automotive	
							Messaged Dorman and they say: 603-001 72oz 6.5"W x 4.75"H x 4"D 4-5/32" center to center on the upper mounting holes
8	1		Coolant Reservoir: Dorman 603-001	\$11.99	\$11.99	https://www.summitracing.com/parts/rnb-603-001	5/16" single barb
9	2		3/4" to 3/4" Barbed Fitting	\$9.99	\$19.98	Amazon.com: CNRAQR 5/8"to 3/4" Hose Barb	Fit between hose from pump and coolant loop hose
10	2		3/4" to 7/8" Hose Adapter	\$9.99	\$19.98	Amazon.com: AC PERFORMANCE ID 7/8" to 3/4"	Pump fittings are 22mm (7/8")
11	1		EBP40 Electric Booster Pump (Davies Craig)		\$0.00	https://daviescraig.com.au/product/ebp40-electric-booster-pump-12v-9040	
12	1	84707	3/4" Manual Adjustment Heater Valve	\$12.99	\$12.99	https://www.summitracing.com/parts/fss-84707?seid=srese2&ppckw=pmax-msn-stock-replacement&utm_source=google&utm_medium=cpc&utm_campaign=2019%20Summer%20Promo	
13	1	SEN-FM18	Coolant Flow Meter with Temperature Sensor	\$68.99	\$68.99	SEN-FM18T10 Coolant Flow Meter with Temperature Sensor, 10K Ohm, Reed Switch	
	0		5/8" to 3/4" Hose adapter			https://www.amazon.com/CNRAQR-Coupler-Fitting-Reducer-Aluminum/dp/B0F62L1J2J/ref=sr_1_15?crid=2R	
			Total		\$941.21		

Pressure Transducer

- [PX3AN2BS100PAAAX Honeywell Sensing and Productivity Solutions | Pressure Sensors, Transducers - Industrial | DigiKey](#)
- X2
- \$129.82



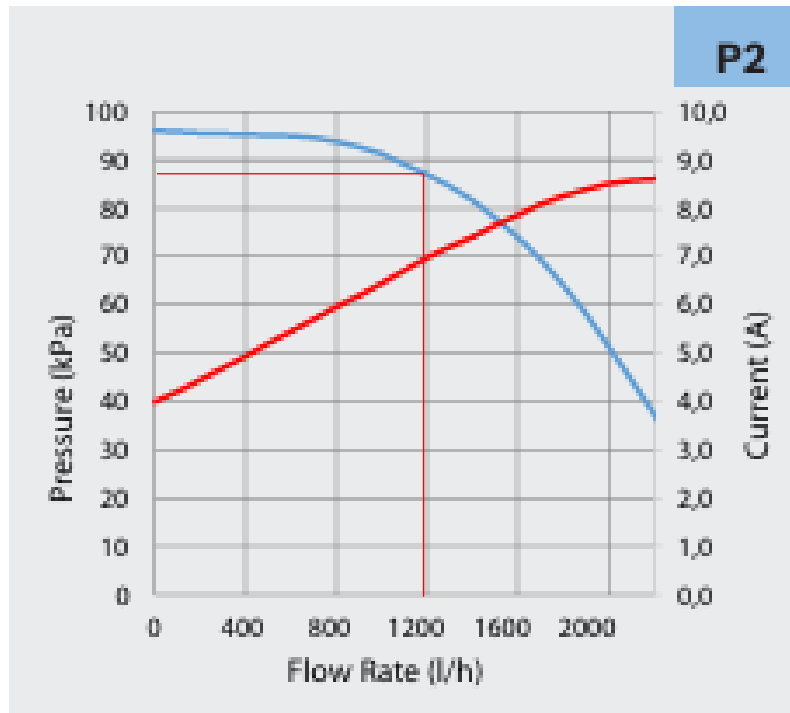
Coolant Reservoir

- [Dorman 603-001 Dorman Coolant Reservoirs | Summit Racing](#)
- \$11.99



Pump

- Max Flow: 20 L/min
- Max Pressure:
 - EDU/Flow meters → 31.395 kPa
 - Heat Exchanger → 7 kPa
 - Total → 38.395 kPa * 1.5 FOS = **58 kPa**



- [Bosch 039202453G Water Cooling Pump, Compact Design | Order Online](#)
- 20 L/min → **87 kPa**



Flow Restrictors

- $\frac{3}{4}$ " - [Four Seasons 84707 Four Seasons Heater Valves | Summit Racing](#)
- Assumptions:
 - Minimum pressure drop: 2-3 kPa
 - Maximum pressure drop: ∞ kPa

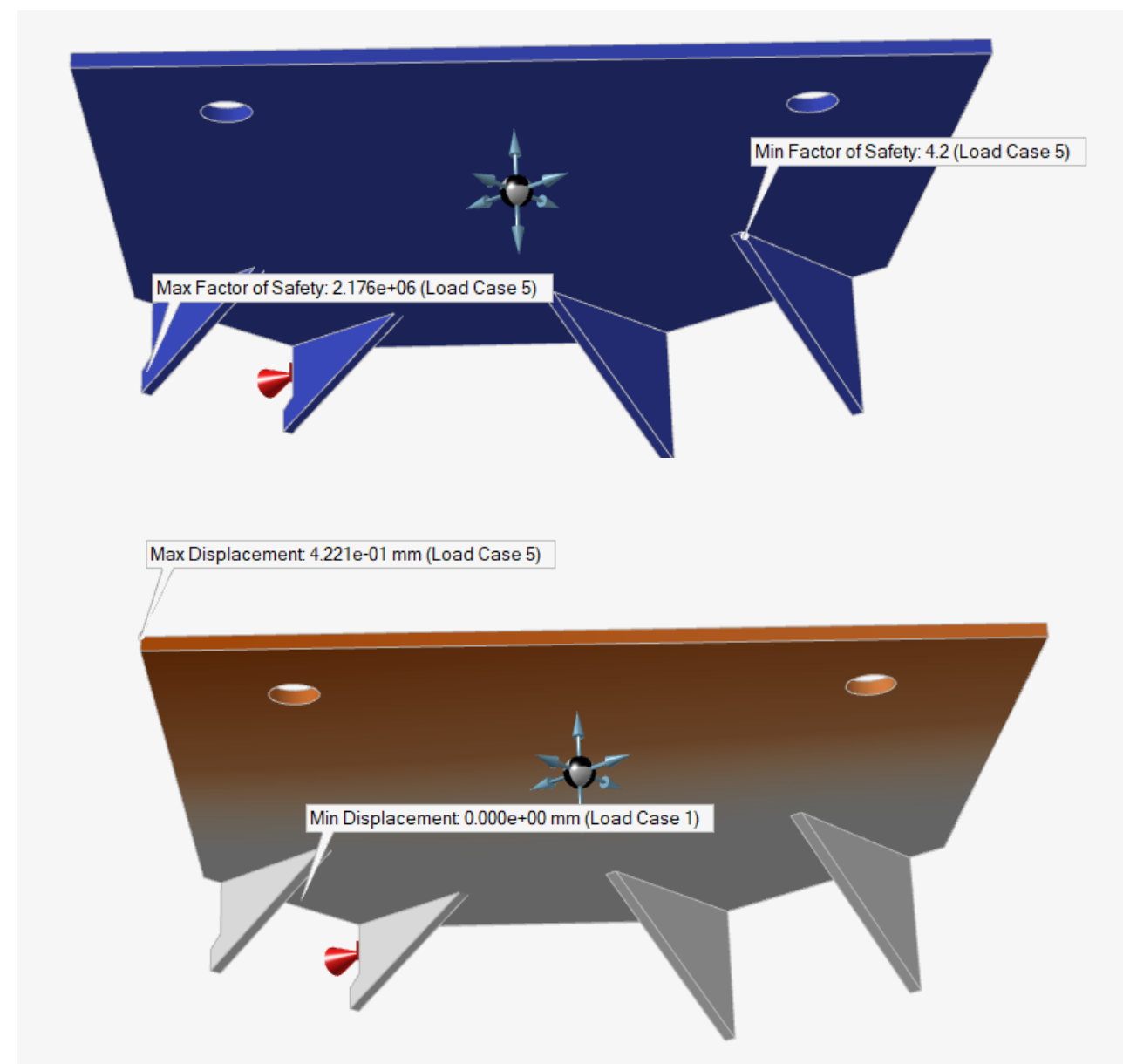
- $\frac{5}{8}$ " - [Four Seasons 84706 Four Seasons Manual Shut-Off Valves | Summit Racing](#)



**Slides on finding location for the HX in the vehicle
were deleted since they show confidential CAD**

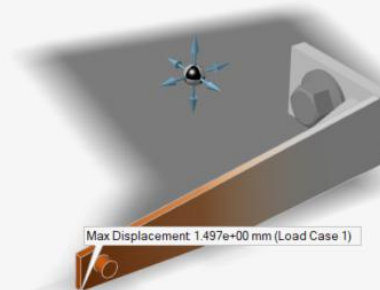
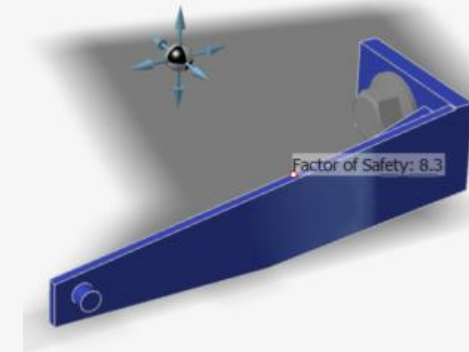
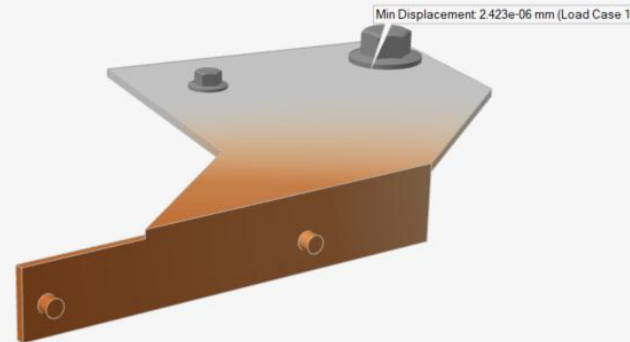
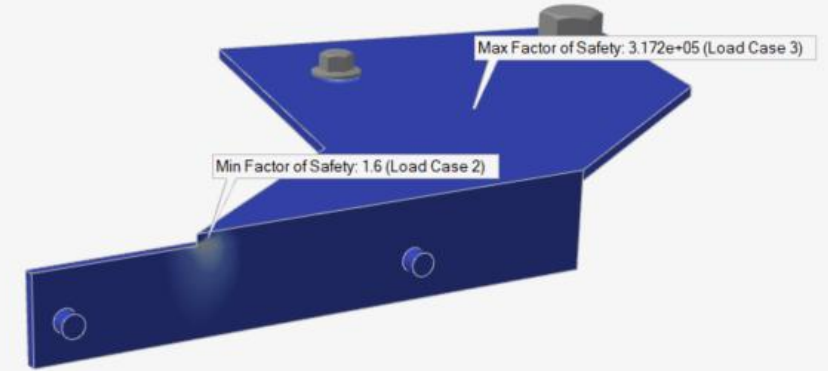
HX Location 1 and Mount

- Material: 4130 Steel
- Thickness: 1/8"
- Loads:
 - 2g vertical
 - 5g longitudinal/lateral
- Mass:
 - $1.2 + 0.11 \times (18 \text{ plates}) = 3.18 \text{ kg}$
 - $0.054 \text{ liters} \times 18 \text{ channels} = 0.972 \text{ liters} = 0.97 \text{ kg}$
 - Total = 4.152 kg



Coolant Reservoir Mount FEA

- Material: 4130 Steel
- Thickness: 1/8"
- Loads:
 - 2g vertical
 - 5g longitudinal/lateral
- Mass of coolant reservoir:
 - 72oz water = 2.125kg



Information Requests

- [General Inquiries | Alfa Laval](#)
- Call [Contact Emmegi Heat Exchangers Inc. | USA](#)
- [About Bell & Gossett Pumps | Contact & Support for Xylem Bell and Gossett Pump Products | Xylem US](#)

Additional Resources

- OG HX dimensions [bell-gossett-brazed-plate-heat-exchanger-brochure.pdf](#)
- OG HX flow and heat rejection [bell-&-gossett-brazed-plate-heat-exchanger-selection-tables.pdf](#)
- HX dimensions [Alfa Laval](#)
- HX Flow and heat rejection table [1 new message](#)
- HX flow and heat rejection graph [WB-Series-Brazed-Plate-Oil-Coolers.pdf](#)
 - Parent: [WB Series Brazed Plate Oil Coolers | Emmegi Inc.](#)
- Heat exchanger selection tool [PHE Select tool | Alfa Laval](#)